

Vehicle Layer

GPS Tracker (Concox GT06S)
primary source — fully automatic
fixed in the bus, no driver needed

Driver Smartphone
automatic fallback
used only if needed

every 5 s
TCP over GSM

REST / HTTPS
(fallback)

Backend (single low-cost server)

Telematics Gateway
(Traccar) decodes
tracker messages

WebSocket
(≈ 1 s)

REST poll
(backup)

Application Server (Node.js)
Source Selection · Data Validation
ETA Calculation · Trip Monitoring
Real-Time Broadcast

*everything runs on
one server — no external
services needed*

PostgreSQL
database: trips,
routes, users

Redis
cache · pub/sub
instant delivery

WebSocket push:
location & ETAs

Client Layer

REST API:
manage & query

Passenger Mobile App

live map · arrival times · alerts
bilingual English/Bangla

Admin Dashboard

routes · schedules · fleet
live operations map

➡ real-time location data

↔ request ↔ response (REST)

Acquisition

- 1 GPS fix captured on the bus**
captured every 5 s while the bus is moving

$t = 0\text{ s}$

- 2 Transmitted and decoded**
sent over the cellular network; decoded in $\approx 1\text{ s}$

$\approx 1\text{ s}$

- 3 Best source selected**
tracker preferred; phone covers tracker outages

- 4 Faulty readings rejected**
inaccurate fixes, impossible jumps and jitter removed

- 5 Arrival times calculated**
remaining distance $\div$ recent speed, per stop ahead

- 6 Stored and published**
saved in PostgreSQL; published through Redis

- 7 Pushed to every subscribed phone**
WebSocket delivers position and new arrival times

Processing

Delivery

Result: passenger's map & ETAs refresh

GPS fix $\rightarrow$ server: median 5.9 s (mean 7.4 s)

5.9 s (median)